scottm59@msu.edu
mackenzie-scott-121697242
0009-0004-7242-0204
kenzerkay

Mackenzie Ticoras

Education

2023 – present **Doctor of Philosophy in Astrophysics and Computational Math, Science and Engineering (dual PhD)**, Michigan State University
Advisor: Brian O'Shea

2023 – 2026 **Masters of Science in Physics**, Michigan State University

2019 – 2023 **Bachelor of Science in Physics and Astronomy (double major)**, The Ohio State University
Honors Research Distinction and Honors in the College of Arts and Sciences
Undergraduate Thesis: "Refinement of Stellar Intensity Interferometry at VERITAS"
Advisor: Michael Lisa

Research Experience

2023 – present **Graduate Researcher**, Michigan State University
Investigate the circumgalactic medium using FOGGIE simulations

2020 – 2023 **Undergraduate Researcher**, The Ohio State University
Stellar measurements with digitized intensity interferometry (SII)

Publications

[1] **Mackenzie Ticoras** et al. "Figuring Out Gas & Galaxies in Enzo (FOGGIE). XV. Examining the Spatial and Kinematic Relationship between Circumgalactic Mg II and O VI". In: *arXiv e-prints*, arXiv:2601.02348 (Jan. 2026), arXiv:2601.02348. DOI: 10.48550/arXiv.2601.02348. arXiv: 2601.02348 [astro-ph.GA].

[2] A. Acharyya et al. "An Angular Diameter Measurement of β UMa via Stellar Intensity Interferometry with the VERITAS Observatory". In: *ApJ* 966.1, 28 (May 2024), p. 28. DOI: 10.3847/1538-4357/ad2b68. arXiv: 2401.01853 [astro-ph.SR]
Corresponding Author: Mackenzie Ticoras.

Conference Presentations

November 2024 **Resolving The CGM and its Impact on Galaxy Evolution**, *"Investigating kinematics and spatial distribution of multiphase gas in the CGM"*, Santa Cruz, Chile

May 2023 **Ohio State Stellar Intensity Interferometry Conference 2023**, *"Measurement of Merak with VERITAS SII"*, Columbus, Ohio

January 2023 **VERITAS Collaboration Meeting Winter 2023**, *"An Angular Diameter Measurement of β UMa"*, Tuscon, Arizona

July 2022 **OSU Astronomy Department Undergraduate Summer Symposium**, *"A Uniform Disk Visibility Curve for bet UMa"*, Columbus, Ohio

June 2022 **240th Meeting of the American Astronomical Society**, *"An intensity interferometry measurement of gamma Orionis (Bellatrix) with the VERITAS telescope array"*, Pasadena, California

April 2022 **Eastern Great Lakes Section of the American Physical Society**, *"An intensity interferometry measurement of gamma Orionis (Bellatrix) with the VERITAS telescope array"*, Youngstown, Ohio

Teaching Experience

2025 – present **Certification in College Teaching (In Progress)**, Michigan State University
This program is designed to help graduate students and postdocs develop their teaching skills specifically with undergraduates in mind.

2021 – 2023 **Student Tutor**, The Ohio State University, Math and Statistics Learning Center
I was a tutor for college-level math classes including college algebra, all levels of calculus, and linear algebra

Scholarships and Fellowships

March 2023 National Science Foundation Graduate Research Fellowship

March 2023 University Distinguished Fellowship

February 2023 College of Natural Sciences Recruiting Fellowship, Declined

April 2022 Undergraduate Research Scholarship at The Ohio State University

March 2022 Goldwater Scholarship

May 2021 Physics Summer Research Scholarship at The Ohio State University

2019-2023 Provost Merit Scholarship (Awarded Semesterly)

Awards and Honors

March 2023 Physics Senior Award at The Ohio State University

April 2022 Best Undergraduate Oral Presentation at EGLS APS Meeting

April 2022 Smith Junior Award at The Ohio State University

2019-2023 The Ohio State UniversityDean's List (Awarded 8 Semesters)

2019-2023 Undergraduate Honors Program member at The Ohio State University

Proposals

2024 **"Probing Galaxy Formation at Low and High Redshifts"**, *ACCESS Allocation*, 4.3M CPU hours
Role: Co-I

Outreach and Mentorship Experience

2024 – Present **Outreach Assistant**, Michigan State University, Michigan State Observatory
Help facilitate outreach events and public nights at the MSU observatory

2023 – Present **Stellar Mentorship Program**, Michigan State University
Mentor 2023 – Present
Program Leader 2024 – Present

2020 – 2023 **Code 4 Community**, The Ohio State University
President 2022 – 2023
Treasurer 2021 – 2022
Member 2020 – 2023

Students Mentored

2025 – present Jazzmin Partirdge

Other Service

2024 and 2025 **Prospective Student Liason**, Michigan State University

2024 – 2025 **Graduate Student Curriculum Committee**, Michigan State University

Computation and Technical Skills

■□□□□ basic knowledge
■■□□□ intermediate knowledge with some project experience
■■■□□ extensive project experience
■■■■□ deepened expert knowledge
■■■■■ expert / specialist

	Level	Skill	Years	Comment
Languages:	■■■■□	Python	5	*Extensive use in both past and current research projects and coursework*
	■■■□□	C++	6	*Extensive use in both past and current research projects and coursework*
	■■■□□	LaTeX	5	*Extensive use in both past and current research projects and coursework*
	■■□□□	HTML/CSS	5	*Casual development of personal website and other projects*
Computational Software/Tools:	■■■□□	Parallel Processing	5	*Significant experience in both past and current research projects, including use of MPI and OpenMP*
	■■■□□	GitHub	5	*Extensive use for version control in both past and current research projects, coursework, and personal projects*

	■■□□□	yt	3	*Experience with this Python package for analyzing and visualizing large volumetric datasets, including FOGGIE CGM simulations*
	■■□□□	Root @ Cern Library	3	*Experience with this library for data analysis and visualization in past research projects*
	■□□□□	Machine Learning	1	*Minimal experience with machine learning techniques, including use of scikit-learn and TensorFlow in coursework and personal projects*
Technical Experience:	■■□□□	Instrumentation	3	*Experience with photomultiplier tubes, digitizers, and other instrumentation used in the VERITAS-SII project*
	■■□□□	Telescope Observations	1	*9 nights of on-site intensity interferometry observations at the VERITAS telescopes*